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ARTICLE

The confined [Bmim][BF₄] ionic liquid flow through graphene oxide nanochannel: A molecular dynamics study

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The ionic liquid (IL) flow in graphene oxide (GO) nanochannel plays a key role in the performance of the IL and GO-based fluidics devices and other chemical separator techniques. Here, we investigate the flow behavior of IL in GO nanochannel *via* molecular dynamics simulations. The quantitative relation between slip velocity and shear stress has been identified, showing that the interfacial friction coefficient can be enhanced by almost sixty times, while slip length is reduced by about three orders of magnitude, with the fraction of hydroxylation in graphene ranging from 0% to 15%. The great change of interfacial properties can be attributed to the structure change of IL layer near GO, which is proved by the detailed analysis on density distribution, charge distribution and radial distribution function. Besides, the viscosity will increase as fraction of hydroxylation because of the partial breaking of coulombic ordering of confined IL. Meanwhile, the hydroxyls have more significant effects on IL flow than water flow in GO nanochannel due to the stronger interaction network in IL/GO interface. In a word, hydroxylation can be a convincing method to regulate the IL flow in nanochannel. The quantitative properties of confined IL in GO nanochannel and its relation with fraction of hydroxylation could deepen the understanding on IL and benefit the application of IL and GO in the fields of chemical engineering and other various nanofluidic devices.

1. Introductions

Liquid flow in nanochannel plays a key role in many important nano-technologies, ranging from graphene oxide (GO) membrane for desalination of sea water^{1, 2} and gas separators³ to graphene supercapacitors for energy storage^{4, 5}. In the practical applications, two-dimensional (2D) materials, such as graphene and GO sheets, prefer to self-assemble in paper-like structures with interlayer or intralayer distance on nanometer scale and construct many nanochannels. Inside the nanochannel, the liquid will form a relative ordered layer near the wall^{6, 7}. Such layered structure makes the property of confined liquid to be different with that of bulk liquid and induces many intriguing phenomena, including ultrafast transport of water molecules in graphene nanochannel^{1, 6, 7}, high ion selectivity in subnanometer carbon nanotube and GO membrane⁸⁻¹⁰, nanolinear behavior such as current rectification in charged conical nanochannels¹¹, and so on. Most prior research on liquid flow in nanochannel was limited to aqueous or molecular solvent, such as water, oil, etc.. The ionic liquid (IL) has attracted much less attention despite the fact that ionic liquid are widely used in gas separators¹²⁻¹⁴,

supercapacitors^{15, 16} and nanofiltration^{11, 17}, etc., whose performance is also dominated by its structure and flow in the nanochannel.

ILs are a new class of materials consists of only cations and anions and existed as liquid near room temperature, and have many unique properties such as wide electrochemical window, low vapor pressure, and nonflammability^{18, 19}. Due to the ionic nature²⁰, the interactions between ILs and graphene or GO are expected to be entirely different from that with conventional molecular liquid such as water. Prior research on IL in nanochannel constructed by other two dimensional materials mainly focused on the structure^{21, 22}, self-diffusion of ions¹, and flow behavior under external electric fields^{23, 24}. For example, Futamura *et al.* showed that the charge distribution of IL in graphene nanochannel and reduced coulombic ordering of confined IL *via* X-ray scattering experiments²². Shin *et al.* revealed that the effect of IL pumping along the carbon nanotube-induced by DC electric field increases as the electrode diameter decreases²⁴. However, limited work has been devoted to understanding about quantitative flow properties of IL confined in graphene or GO nanochannel, such as viscosity η , solid-liquid friction coefficient λ , slip length l_s , etc., which has restricted the development of the IL-based applications. A deep and systemic understanding of confined IL flow and the means of controlling IL transport in GO nanochannel is highly desired in practice.

In this work, we systemically study the flow behavior and properties of IL inside graphene and GO nanochannel *via* the combination of equilibrium and non-equilibrium molecular

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dynamics (MD) simulations and find that hydroxylation can serve as the effective means to regulate the IL flow. Firstly, the quantitative relations between shear stress and slip velocity are proposed and the relations between interfacial friction coefficient and fraction of hydroxylation (c) are identified. Then the interfacial structure, charge distribution, density distribution, and radial distribution function (RDF) are quantitatively discussed to gain a deep understanding of the molecular mechanism and physics of IL flow. Then, the viscosity of IL in GO nanochannel with different fraction of hydroxylation is summarized, and the underlying mechanism of the change of viscosity is analyzed. Finally, the effect of hydroxyls on the slip length and the difference of IL/GO and water/GO interface is discussed. Our study offers novel insights into the design of nanochannel and control of IL flow behavior, which could benefit the application of GO and IL in the super capacitor or other chemical separator techniques.

2. Models and Methods

All the simulations in this work are performed using large-scale atomic/molecular massively parallel simulator (LAMMPS)²⁵. The timestep for integrating Newtonian equations of motion is 1.0 fs, which is validated to ensure the energy conservation. IL has many different kinds of cation and anion, we choose the most typical one, 1-Butyl-3-methylimidazolium tetrafluoroborate, $[\text{Bmim}]^+ [\text{BF}_4]^-$ in the present work due to its successful utilization in many electrochemical devices such as super capacitor, lithium battery^{26, 27}. The atomic structure of $[\text{Bmim}]^+ [\text{BF}_4]^-$ is shown in Fig. 1(a). The atomic structure of GO composed of many different functional groups on the graphene layer such as hydroxyl, epoxy, and carbonyl and so on^{28, 29}, where hydroxyl groups remain rich and towards a long-living quasi-equilibrium state. Hence, we only consider the hydroxyls groups in the structure of GO in this work, and the fraction of hydroxylation c is defined as: $c = n_{\text{OH}}/n_{\text{C}}$, where n_{OH} and n_{C} are the number of hydroxyls and carbon atoms, respectively. The hydroxyl will be randomly added to the two

sides of graphene sheet. The size of GO sheet is $10.12 \times 5.01 \text{ nm}^2$, and a periodic boundary condition (PBC) is applied in the x - and y -direction while an open boundary is used in the z -direction. To construct the atomic model, we firstly place 400 pairs of ions in the GO bilayer with an interlayer distance $d = 4.0 \text{ nm}$, then we move the GO sheet and decrease the interlayer distance d and track the evolution of pressure of confined IL. From the pressure-distance relation shown in Fig. S1, we can see that the pressure will be $\sim 0 \text{ bar}$ when $d = 2.8 \text{ nm}$. Hence, we set $d = 2.8 \text{ nm}$ in our simulations to explore the structure and flow behavior of confined IL.

The non-polarizable OPLS-AA potential was employed for the IL³⁰ and GO³¹. The parameters of van der Waals interactions, electrostatic interactions, bond, angle, and dihedrals in IL and GO have been used successfully to capture the liquid property of IL and the structural, mechanical and thermal properties of GO^{3, 29, 32, 33}. The $-\text{CH}_3$ and $-\text{CH}_2-$ groups in $[\text{Bmim}]^+$ are modelled as united atoms (UA). The UA model is a good approximation in this study, since the contribution of the high-frequency vibrations of hydrogen atoms to flow transport can be negligible³⁴. The partial charge of atoms in GO and IL are from Jorgensen et al.'s and Liu et al.'s work^{30, 31}, respectively, which are also summarized in Table S1-2. The IL-GO interactions included both van der Waals and electrostatic terms. The former one is described using the Lennard-Jones potential function and the Lorentz-Berthelot mixing rules are used to model the parameters, which are truncated at 1.2 nm. The later one, long-range coulombic interaction, is computed using the particle-particle-particle-mesh (PPPM) algorithm.³⁵ The SHAKE algorithm are also used to reduce high-frequency vibrations of hydrogen in GO³⁶. In the simulations, IL ions and hydroxyls are kept at a constant temperature of 300 K using Berendsen thermostat³⁷, with the temperature calculated after removing the center-of-mass velocity. The carbon atoms in GO sheets are fixed in the MD simulations to maintain the planar structure of the GO sheet considering GO stacked in a thin-film case in the electrode materials or the paper like membrane.

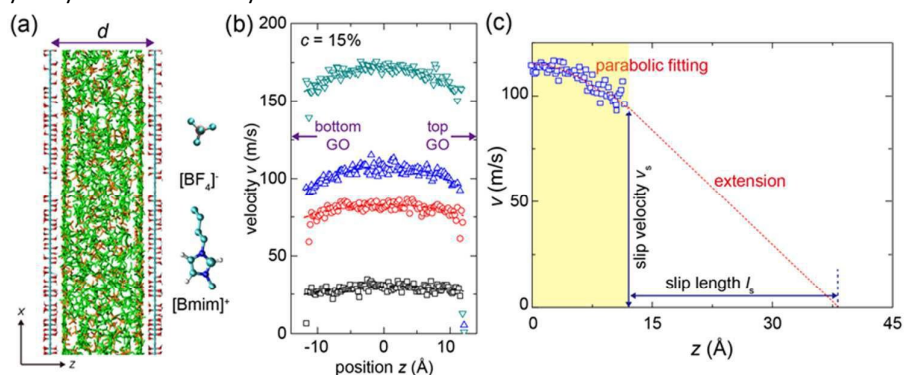


Fig. 1 (a) The illustration of GO nanochannel with $d = 2.8 \text{ nm}$ and structure of IL, $[\text{Bmim}]^+ [\text{BF}_4]^-$. For IL, the green and orange color represent $[\text{Bmim}]^+$ and $[\text{BF}_4]^-$, respectively. For GO, the cyan, red and white color represent carbon, oxygen and hydrogen atom, respectively. (b) The parabolic velocity profiles of IL along the flow direction obtained from MD simulations, which meets the

Poiseuille flow well. (c) The definition of slip length l_s and slip velocity v_s in the MD simulations, where the red dash line represents the parabolic fitting of the velocity profile.

In our non-equilibrium MD (NEMD) simulations, the external force is added on the atoms in IL to produce a pressure-driven Poiseuille flow. Different external force will produce flow with different velocity, and it usually takes about several hundreds of picoseconds to reach a steady flow (balance between external force and friction from GO). After the steady flow is achieved, MD simulation continues to run additional 1 ns to collect the data. Recent studies indicate that thermostat could have a significant effect in the dynamics of pressure-driven confined liquid flow, when the flow is strong or the Reynolds number (Re) is high (10^3 - 10^5)³⁸⁻⁴⁰. To measure this effect, we calculate the Reynolds number for flow in our model firstly, showing a very low one ($Re = \sim 0.019$), and then the distribution of temperature in flow is calculated and shown in **Fig. S2**, indicating a profile-unbiased temperature for our simulation. The low Reynolds number and profile-unbiased temperature implying that NEMD simulations can be used to explore the flow property in GO nanochannel⁴⁰, which have also been implemented well in the studies on liquid flow in nanochannel^{41, 42}. Via applying different external pressure, we obtain the flow with velocities from 0 to 200 m/s as shown in **Fig. 1(b)**. The interfacial friction shear stress between IL and GO can be calculated from the balance between friction and external force as $\tau = Nf / 2S$, where N is the number of IL atoms, f is the external force on every atom, S is the area of the GO sheet. Due to the shear stress between IL and GO, the velocity profile of IL can be fitted well using a parabolic line as shown in **Fig. 1(b)**, which is also the characteristic velocity of the Poiseuille flow.

3. Results and Discussions

3.1 Interfacial slippage of IL in GO nanochannel

As shown in **Fig. 1(b)**, there exist obviously interfacial slippage between IL and GO sheets. To explore the interfacial slippage quantitatively, we calculate the solid-liquid friction coefficient λ between IL and GO sheets from MD simulations, which is the physically relevant property to describe the interfacial dynamics⁴³. As a dissipation coefficient, λ can be calculated from the linear response theory from equilibrium molecular dynamics (EMD) simulations via the Green-Kubo relationship, which relates λ to the autocorrelation function of fluctuating shear forces between IL and GO sheets at equilibrium⁴⁴,

$$\lambda = \frac{1}{Sk_B T} \int_0^{\infty} \langle F_x(t) F_x(0) \rangle dt \quad (\text{Eq. 1})$$

Where S is the area of IL/GO interface, T is the temperature of IL, k_B is the Boltzmann constant, F_x is the time-dependent total shear force acting on the confined IL, x is the direction of the flow and the angular bracket represents the ensemble average, namely the shear force autocorrelation function (ACF)⁴⁵. From the ACF of shear force and λ - t relations shown in

Fig. S3, the convergence time is on the scale of 500 ps for this studied system. So we run 10 independent 2 ns long simulations to obtain the convergent friction coefficient. From the results in **Fig. 2(b)**, we can see that λ will increase greatly as c increases in an almost linear relationship. To reveal the intrinsic property of the fluid-solid interface⁴⁶, we further performed NEMD simulations.

In the NEMD simulations, we calculate the interfacial shear force directly and transformed it to interfacial shear stress τ . As shown in **Fig. 2(a)**, τ is summarized as a function of v_s in GO channel with different c , where v_s is slip velocity and can be obtained from the velocity profile of IL flow. From the results of τ and v_s , the fact is that τ will increase gradually as v_s increases and they can be fitted well using an inverse hyperbolic sine (IHS) relation as $\tau = \tau_0 \sinh(v_s/v_0)$. The IHS relation is derived from the transistor state theory model of Yang⁴⁷, where the slip flow is described as a collective thermal diffusion of liquid atoms over the periodic solid wall and which has been used successfully to quantify the friction coefficient of water in graphene and carbon nanotube nanochannel^{42, 48}. From the IHS fitting, we can derive two parameters: τ_0 and v_0 as summarized in **Table 1**, which can be used to calculate the interfacial friction coefficient via $\lambda \approx \tau_0/v_0$ (slopes of the τ - v_s curves at low slip velocity in **Fig. 2(a)**).

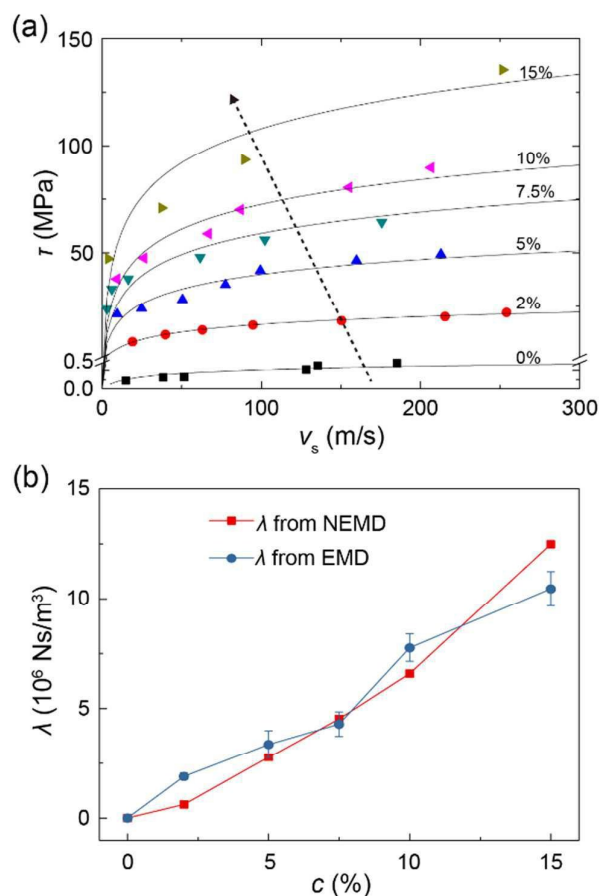


Fig. 2 (a) Friction shear stress τ as a function of the slip velocity at the ionic liquid/GO interface with c ranging from 0% to 15%, where solid lines represent the IHS relations. (b) The friction coefficient λ as a function of c from EMD and NEMD simulations.

Table 1. The parameters τ_0 and v_0 derived from the IHS fitting of the τ - v_s curves

c (%)	0.0	2.0	5.0	7.5	10.0	15.0
τ_0 (MPa)	0.227	5.180	9.947	14.222	15.692	23.055
v_0 (m/s)	19.829	8.174	3.574	3.133	2.515	1.847
$\lambda \approx \tau_0/v_0$ (10^6 N s/m ³)	0.011	0.633	2.783	4.538	6.600	12.482

As c increases, the shear stress will also increase to realize the same flow velocity, indicating that a larger friction is induced via hydroxyls. To quantitatively describe the relation, the solid-liquid friction coefficient λ from NEMD as a function of c is summarized in **Fig. 2(b)**. From the results, we can see that λ will increase as c increases and λ from EMD and NEMD simulations agrees well with each other, indicating that the rationality of our results. For $c = 0\%$ or graphene case, $\lambda = 1.1 \times 10^4$ N s/m³, which is close to the friction coefficient of water and graphene (1.48×10^4 N s/m³)⁴². Such small friction should result from the bare graphene and IL layer structure shown in **Fig. 4(a)**, which will be discussed in the following text. However, even adding 2% hydroxyls on the graphene sheet, the friction coefficient will be enhanced almost sixty times and up to 63.30×10^4 N s/m³, indicating that hydroxyls have dramatic effect on the friction coefficient due to the strong interaction between hydroxyl and IL fluids. Based on the above discussions, we can find that the interfacial friction coefficient λ is very sensitive to the hydroxylation of GO sheet, suggesting that hydroxylation can

serve as an effective method to tune the flow of IL in GO nanochannel. To obtain a deep understanding on how the hydroxyl functional groups influence the molecular structure and the behavior of flow, we will analyze the interfacial structure of IL in the next section.

3.2 Molecular structure of IL-GO interfaces

As we all know, the molecular structure of IL near interface has a significant effect on the slip flow in the nanochannel. Several researchers have studied the interfacial structure of water in graphene, GO and CNT nanochannel. The results showed that there exist an ordered water layer near the surface, which can greatly influence the transport of water in the nanochannels^{41, 42, 48, 49}. To investigate the distribution of IL in the nanochannel, we firstly calculate the spatial density distribution in GO nanochannel with d ranging from 7.5 Å to 20.0 Å. As shown in **Fig. 3(a)**, we can see that IL can form stable monolayer, bilayer and trilayer structures in the GO channel. For d less than 11.6 Å, an IL monolayer structure can be observed obviously. As d increases, the IL will form bilayer and trilayer structure when h

less than 15.7 Å and 18.8 Å, respectively. With increasing interlayer distance continuously, the IL cannot remain the trilayer structure and will become to the structure similar to bulk IL case, which is like case of water in GO nanochannels⁴¹. The structure of IL in nanochannel is largely determined by the balance among van der Waals interactions, hydrogen bonding, electrostatic force and the geometrical property of both IL and hydroxyl in GO. The typical structure of monolayer, bilayer and trilayer are shown in Fig. 3(b), which clearly shows the layered IL structure.

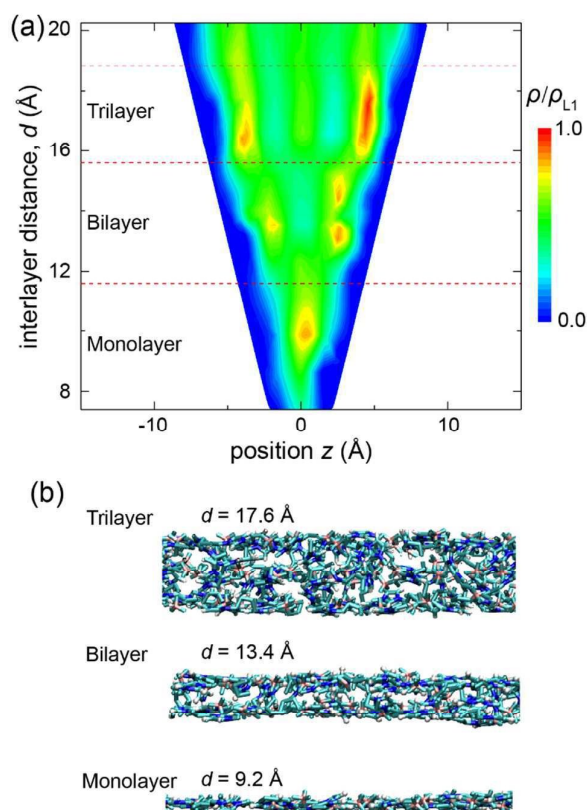


Fig. 3 (a) Contour plot of the IL density distribution $\rho(z, d)$ between GO at $c = 10\%$ with a reference to the density of the first IL layer near GO sheet ρ_{L1} , where d is the interlayer distance between two GO sheets. The color bar in the plot represents ρ/ρ_{L1} . (b) The structures of IL layer in the GO nanochannels with different d for the monolayer, bilayer and trilayer IL, respectively.

To capture the feature of IL flow in nanochannel, the interlayer distance of GO sheets is 28 Å in the present work, which is indeed large enough to make sure the property of center IL close to the bulk IL. Firstly, we plot the density distribution as a function of z position in the graphene nanochannel as shown in Fig. 4(a). The contributions from $[\text{Bmim}]^+$ and $[\text{BF}_4]^-$ to the total density are also inserted into the Figure. From density distribution, we can see that there exist obviously two peaks near GO sheet, namely peak 1 and peak 2, which has a higher density comparing with bulk case, while the center part ($z >$

7.5 Å) have a density similar to the bulk case. Comparing with $[\text{Bmim}]^+$, $[\text{BF}_4]^-$ is closer to GO sheet indicating that it has a stronger interaction with GO. To see the role of hydroxyls, we also summarized the density distribution of IL in nanochannel with different c in Fig. 4(b). From the results, we can see that the density of peak 1 will decrease as c increases, while that of peak 2 increases, indicating that some ions will depart from the GO sheet due to the interaction between IL and hydroxyls. Secondly, the charge distribution of IL across the nanochannel is also summarized in Fig. 4(a-b), showing that there exist obviously charge layers near the GO sheets. A negative charge layer ($\sim -0.008 e$) appears at the distance 3.3 Å to GO, following a positive charge layer ($\sim 0.009 e$) at the distance 4.0 Å, and a negative charge layer ($\sim -0.007 e$) at the distance 5.18 Å. By analyzing the elementary contribution to total density, we can know that these three charge layers are mainly contributed by $[\text{BF}_4]^-$, imidazole ring in $[\text{Bmim}]^+$, $[\text{BF}_4]^-$, respectively. As we all know, the bulk IL is electrical neutrality due to the high coulombic ordering. However, the nanoconfinement will partial break the coulombic ordering and induce to the charge layer as shown in Fig. 4(b). The partial breaking of coulombic ordering is very important to the property of IL in nanochannel and will be discussed in the next section. In a word, the distribution of density and charge across the channel is largely determined by the interactions between GO and IL, which can be tuned by changing fraction of hydroxylation. It has to be mentioned that there exist an interesting key point: the peak position of density and charge distribution almost does not change as c increases. That's to say, intercalating hydroxyls between graphene and ILs does not change the equilibrium distance between ILs and GO sheets due to the strong interaction between cations, anions and hydroxyls and van der Waals interaction, resulting a similar distance as shown in Fig. 4(d). This is very meaningful because the ILs flow through GO nanochannel with a partial oxidation pattern can be regarded as a continuum flow without changing the geometry structure but only modifying the boundary conditions.

To capture the molecular insights of first IL layer (defining as distance between IL and GO less than 6 Å), we plot the two-dimensional RDF between $[\text{Bmim}]^+$ and $[\text{BF}_4]^-$ $g_{C-A}(r)$ in Fig. 4(c). The results show that the RDFs of ILs in GO nanochannel with different c (0%, 10% and 15%) almost overlap with each other, indicating that hydroxyls have a negligible effect on the first IL layer. Taking the RDFs of IL in bulk case as a comparison, and found that the position of first IL peak is almost the same to that of IL bulk, suggesting that the average cation-anion interatomic distance of the neighbored ions is unaffected in the GO nanochannel, i.e., $d_{C-A} = 4.17$ Å. From the snapshots of IL-GO interface in Fig. 4(d), the imidazole ring of cation prefers to lay on the GO sheet in parallel, showing a strong π - π convergent interactions and leading to arrange more IL ions in the neighbored space of GO sheets. Thus, the nanoconfinement mainly leads to more compact IL ions in the first liquid layer than in the bulk case while maintaining the same interionic distance. Due to the IL flow critically depends on the first liquid layer near GO sheets, which will provide a

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modified boundary condition and different IL property for the fluid, flow features that beyond the bulk IL would be expected.

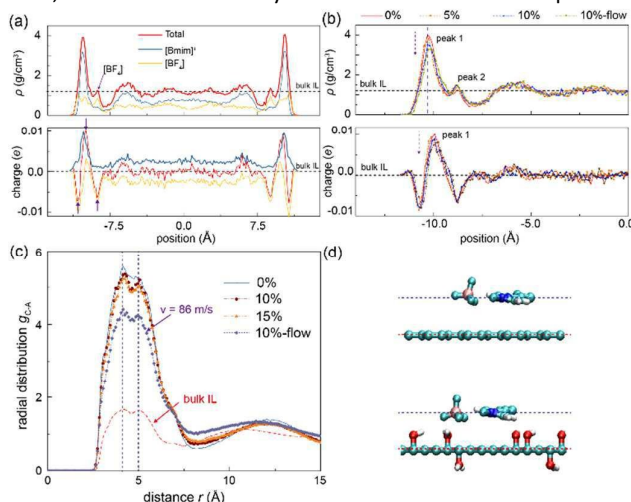


Fig. 4 (a) The density and charge distribution of IL in graphene nanochannel, where the contributions from [Bmim]⁺ and [BF₄]⁻ are also showed. (b) The density and charge distribution of IL across the GO nanochannel with different c . The sharp density peak approaching the GO represents the first IL layer. Insert plot is the peak value of the first peak of density profiles varies with c . (c) RDF g_{CA} of cations and anions in different GO nanochannel, which shows that the position of peak does not change as the GO nanochannel changes. (d) The interfacial structure of [Bmim]⁺, [BF₄]⁻, hydroxyl and graphene, indicating that the similar equilibrium distance between IL-graphene and IL-GO interfaces.

3.3 The viscous nature of IL flow

In the above section, the molecular structure, density distribution and charge distribution of IL in nanochannel have been discussed. Besides the nanoconfinement, the hydroxyls can also influence the structure of IL. To quantitatively describe the change induced by hydroxyls, we plotted the peak value of density and charge distribution (ρ_{den} and ρ_{charge}) against c in **Fig. 5(a)**. As c increases from 0 to 15%, ρ_{den} and ρ_{charge} decrease 22.3% and 16.0%, respectively. That's to say, the hydroxyls can result in small change on the structure of IL in nanochannel. However, such a small change of IL's structure can have significant effect on its self-diffuse coefficient and flow behavior. As shown in **Fig. S5**, we plot 2D diffuse coefficient D_{xy} as a function of the fraction of hydroxylation. We can see that D_{xy} decreases gradually as c increases due to the strong interaction between hydroxyls and ILs. Simultaneously, the solid-liquid friction coefficient increases almost three orders of magnitude as c verifying from 0 to 15%. As a result, the curvature of velocity profile increases obviously as c increases, changing from flat to parabolic curve as shown in **Fig. 5 (b)**, and the trend is even more significant as c increases.

Besides the flow behavior, the hydroxyls can also modify the viscosity of IL in nanochannel, which is one of important factor to describe the IL flow. To capture the effect of hydroxyls on

the viscosity, the viscosity of IL η is calculated using Green-Kubo equation based on the linear-response theory,

$$\eta = \frac{V}{k_B T} \int_0^{\tau_c} \langle P_{xy}(\tau) P_{xy}(0) \rangle d\tau \quad (\text{Eq. 2})$$

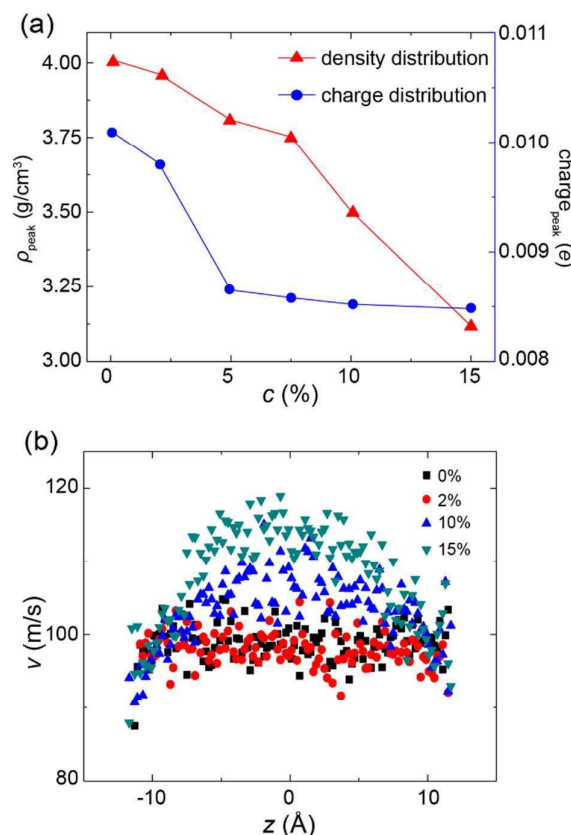


Fig. 5 (a) Peak value of density and charge distribution, ρ_{den} and ρ_{charge} , as a function of fraction of hydroxylation. (b) Flow velocity profiles of IL at different c on the graphene sheet with $h = 2.8$ nm.

Where V is the volume of IL and P_{xy} is the shear pressure at equilibrium. Based on the simulation and ACF of pressure shown in **Fig. S4**, we find that η will increase as τ_c and will up to convergence at ~ 300 ps, so we run 10 independent 2 ns long simulations to calculate η as done in the calculation of friction coefficient above.

The calculated viscosities of IL in different GO nanochannel are summarized in **Fig. 6(a)**. As denoted by the purple dash line, the viscosity of bulk IL $\eta = 108.94 \pm 8.83$ mPa·s at 300 K, which has small difference comparing with the experimental value of bulk [Bmim]⁺[BF₄]⁻ (110.308 mPa·s)⁵⁰. However, when the IL is placed in a graphene nanochannel, η will decrease significantly and lower to 43.89 ± 2.32 mPa·s. Such a significant decrease can be attributed to the partial breaking of coulombic ordering of IL. As shown in **Fig. 4(b)**, the charge distribution of IL will form several layers structure and then reduce the coulombic ordering. In the recent experimental work, *Futamura et al.* also studied the charge distribution of IL in bilayer graphene nanoconfinement via the X-ray scattering experiments and found that the nanoconfinement can indeed reduce the coulombic ordering greatly²². On the other hand, the partial

breaking of coulombic ordering can reduce the electronic interaction within IL effectively, which will lead to the decrease of viscosity of IL. It should be noted that the interlayer distance also has a large effect on the viscosity of IL and flow behavior.

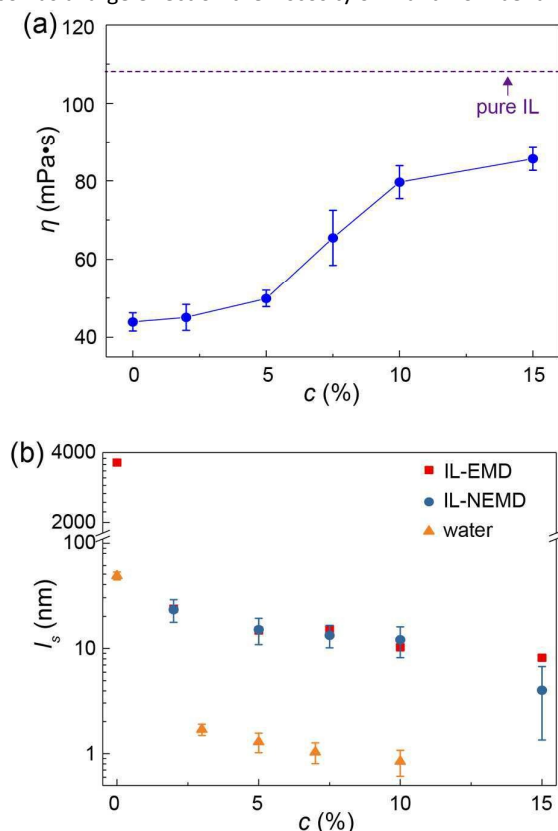


Fig. 6 (a) The viscosity of IL in the GO nanochannel as a function of the fraction of hydroxylation. The dash line represents the viscosity of bulk IL as a comparison. (b) The slip length plotted against c for GO nanochannel. The circle symbols represent the values of l_s from NEMD simulation by fitting the velocity profile and the square symbols are calculated by the Green-Kubo formula in EMD simulation. As a comparison, we also plotted l_s of water-GO interface from the reference⁴¹.

As shown in Fig. 3(a), when the interlayer distance is small ($h < 15$ Å), IL will form to monolayer or bilayer structure, whose property will be close to the crystal IL and should have a larger viscosity comparing with the bulk case. When the interlayer distance is very large, the GO will have slight effect on the structure of IL and the property of IL will be close to that of bulk IL.

However, the hydroxyls on the graphene sheets will have a different effect on the viscosity of IL. As shown in Fig. 6(a), the viscosity increases from 43.89 to 80.2 mPa·s gradually when c increases from 0% to 15%. This enhancement of viscosity can also attribute to the change of coulombic ordering. When the hydroxyls added to the graphene sheets, it will influence the charge distribution of IL and reduce the peak value as shown in Fig. 4(b) and Fig. 5(a). The reduction of peak value of charge distribution will weaken the breaking level of coulombic

ordering and enhance the electronic interactions within IL. As a result, the existence of hydroxyls will enhance the viscosity of IL in nanoconfinement. Under the two competitive mechanisms based on nanoconfinement and hydroxylation, the viscosity will increase as c and approach the bulk case gradually. That's to say, the hydroxylation combining with nanoconfinement can be a convincing tool to control the viscosity of IL in GO nanochannel.

3.4 The hydroxyl-dependent slip length

Furthermore, the hydroxyls can modify the slip length l_s of IL flow in GO nanochannel, which can be obtained by the viscosity via $l_s = \eta/\lambda$ ⁴³, here η and λ are the viscosity and friction coefficient from the simulations. l_s is showed as a function of c represented by solid circle symbols in Fig. 6(b). It should be pointed out that the relation $l_s = \eta/\lambda$ was critiqued since it cannot describe the intrinsic property of solid/fluid interface well⁵¹. Hence, we also calculate the interfacial slip length according to the definition of Navier slip length $l_s = v_s/(dv/dz)|_{z=z_{\text{wall}}}$, where dv/dz is the slope rate of the velocity profile at the position of IL/GO interface as shown in Fig. 1(c), z_{wall} is the position of wall and is defined as the average position of IL near the GO sheets. The blue square symbols in Fig. 6(b) represent l_s from NEMD results, which are consistent with that from EMD results. Our results suggest that the EMD can capture the essential fluid property in the GO nanochannel. For graphene state ($c = 0\%$), the slip length is about ~ 4 μm , which is far large than the interlayer distance 2.8 nm and lead into a pluglike velocity profile displaced in Fig. 5(a). When c is up to 10%, the slip length decreased dramatically to ~ 12 nm due to the strong interaction between IL and GO sheets and lead into a parabolic velocity profile displaced in Fig. 5(a). As a comparison, we also summarize l_s of water-GO interface from the recent work into the same Figure⁴¹. We can see that l_s of water in nanochannel with 10% hydroxyls was decreased to 1.7% of that in graphene nanochannel, while l_s of IL will be decreased to 0.2%. Such a big difference between confined water and IL should originate from the ionic nature of IL. It can be expected that there exist a strong interaction between ions and hydroxyls due to the modified hydrogen bond coupling with electrostatic interaction, such stronger interaction in IL was called as Z-bond⁵². The Z-bond network between IL and hydroxyl is stronger than hydrogen bond network formed by water and hydroxyls. Consequently, the hydroxyls can have more significant effect on IL flow in GO nanochannel. Hence, exist of hydroxyls can heavily restrained the slip flow of IL in GO nanochannel. Meanwhile, the variation of slip length as fraction of hydroxylation also corresponds to the change of interfacial friction coefficient and viscosity of IL, which reflects that the reasonableness of our simulation methods.

Conclusions

In summary, we have systemically studied the flow behavior and property of confined IL in the GO nanochannel using MD simulations. The interfacial friction coefficient and slip length

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of confined IL is relatively sensitive to the hydroxyls on graphene sheet even at a fraction of 2%. The quantitative relation between interfacial friction coefficient (or slip length) and fraction of hydroxylation is obtained. The relations are further discussed by analyzing the density distribution, charge distribution, and RDF of confined IL in the GO nanochannel with different c , indicating that the change of structure of first dense IL layer close to GO wall dominate the property of IL. Meanwhile, the hydroxyls have more significant effect on IL flow than water flow in GO nanochannel due to the stronger interaction network in IL/GO interface. This has important practical implications for the design of graphene based nanofluidics devices since hydroxylation is an effective means to regulate the properties of confined ILs. Overall, our work provides novel insights and understanding of the molecular mechanisms for the regulation of liquid flow through nanochannel, which will be invaluable and meaningful for the design and manage of the IL and GO-based electrochemical devices and can be extended to other 2D materials or other chemical separator techniques.

Conflicts of interest

There are no conflicts to declare.

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